

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

- A Denial of service (DoS) attack, specifically a TCP SYN flood, targeting the web server's HTTPS (port 443) service

The logs show that:

- An unfamiliar IP address 203.0.113.0, is sending an abnormally high volume of rapid TCP SYN requests to the web server (192.0.2.1) from source port 54770 without ever completing the 3-way handshake. As a consequence of resource exhaustion, legitimate users experience connection timeouts and receive premature connection resets ([RST, ACK]).

This event could be:

- A targeted, volumetric Denial of Service (DoS) attack aimed at disrupting business operations by completely compromising system availability.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The client sends a TCP packet with the SYN flag set to the server to initiate a connection and synchronize sequence numbers.
2. The server responds with a SYN-ACK packet, confirming the request and allocating system resources for the pending session.
3. The client sends an ACK packet back to the server. Once received, the TCP connection is officially established, and application data can be transmitted.

Explain what happens when a malicious actor sends a large number of SYN packets all at once:

- When an attacker floods the server with rapid SYN packets without sending the corresponding ACK responses, the server allocates memory and keeps

each connection in a "half-open" state while waiting for the handshake to complete. This rapidly exhausts the server's TCP connection backlog queue (memory buffer). Once the backlog table is completely saturated, the server can no longer process or accept any new legitimate incoming connection requests.

Explain what the logs indicate and how that affects the server:

- The packet capture reveals that genuine user sessions (such as 198.51.100.23 and 198.51.100.14) initially succeeded in loading /sales.html. However, as 203.0.113.0 sustained its SYN flood, the server became completely overwhelmed. This caused client 198.51.100.5 to encounter an HTTP 504 Gateway Time-out, while the server actively rejected other legitimate requests (198.51.100.16, 198.51.100.7, 198.51.100.22) by issuing forced [RST, ACK] reset packets. Consequently, internal employees and external customers are entirely blocked from accessing holiday sales promotions.